Singapore Institute of Technology
Information and Communications Technology Cluster
AAI4001 Capstone Project

Capstone Project Details (Form B)

A. General Capstone Project Information

Project Information:

(Project Title): ITSM Incident Ticket Classification using Natural Language Processing

(Organization): Panasonic Information Systems Company Asia Pacific

(Initiation):

☒ Project initiated by industry
☐ Project initiated by student

(Multi-disciplinary):

☐ Project is multi-disciplinary (e.g. ICT and ENG)
☒ Project is single-disciplinary (e.g. ICT only)

Candidate Particulars:

(Name of Candidate): Donovan Leong Jia Le

(Matriculation Number): 2302674

(Programme):

Applied Artificial Intelligence

(Project Period): 13/10/2025 to 10/4/2026

Main Academic Supervisor Details:

(Name): Bogdan Cautis

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(Contact Number): 8229 0321

(Designation): Associate Professor

Industry Supervisor Details:

(Name): Praveen Reddy Satti

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(Contact Number): 96501275

(Designation): SAP Application Delivery Management Lead, Application Managed Service

(Department / Division): Managed Services ADM/AMS, PISCAP

B. Capstone Project Details

Project Overview:
(Please use a separate sheet if necessary)

Please provide a description of the candidate's capstone project, including its purpose, etc.

This project aims to enhance IT Service Management (ITSM) by developing an intelligent ticket classification and linking system using Natural Language Processing (NLP). The core focus is on comparing different text embedding models to understand the trade-offs between model size, embedding dimensions, inference speed, and classification accuracy.
The project will implement a production-ready data pipeline that integrates with ServiceNow via API: extracting incident tickets, processing them through custom NLP models, and updating ServiceNow with classification results and ticket linkages.

Four embedding models will be evaluated across a spectrum of size and performance characteristics:

- **all-MiniLM-L6-v2** (384 dimensions, 6 layers, ~80MB) - optimized for speed and high-throughput production deployment
- **Gamma-embedding-300M** (300 million parameters, ~229MB) - balanced performance for general use cases
- **all-mpnet-base-v2** (768 dimensions, 12 layers, ~420MB) - quality-focused baseline with proven reliability
- **Qwen3-embedding-8B** (8 billion parameters, ~4.68GB) - maximum accuracy for scenarios where quality is paramount

While the classification and dashboard components are core deliverables, the project also **explores the feasibility of uncovering deeper relationships** between tickets---such as when one issue may contribute to or stem from another (potential causal links). These insights could help consultants not only identify duplicate or related issues but also trace patterns that point toward common root causes.

Deployment as Vendor-Independent Solution

The final deliverable will be packaged as a containerized application deployable on local infrastructure or cloud-hosted container platforms (AWS ECS, Azure Container Instances, Google Cloud Run, Kubernetes). This architecture provides organizations with a viable alternative to ServiceNow's proprietary Predictive Intelligence, enabling them to harness advanced AI capabilities while maintaining control over deployment infrastructure, avoiding vendor lock-in, and eliminating recurring AI licensing costs. The solution integrates with ServiceNow via standard REST APIs, functioning as an independent microservice that enhances ServiceNow's capabilities without dependency on its AI ecosystem.

Project Objectives:
(Please use a separate sheet if necessary)

Please provide a detailed description of the project objectives, i.e., what is the candidate expected to accomplish

This project aims to develop and evaluate an intelligent ticket classification system that balances accuracy with operational efficiency. The objectives encompass technical implementation, rigorous evaluation, and practical deployment considerations.

The specific objectives are:

- **Design and implement a ServiceNow API integration pipeline** for automated ticket data extraction and result updates, creating a seamless workflow between the ITSM platform and the classification system.
- **Develop and fine-tune four embedding models** (all-MiniLM-L6-v2, Gamma-embedding-300M, all-mpnet-base-v2, Qwen3-embedding-8B) for ticket classification and similarity-based linking. These models represent different points along the spectrum of computational efficiency and accuracy
- **Conduct comprehensive performance evaluation** comparing classification accuracy across ticket categories, linking precision and recall for related ticket identification, inference speed measured in tickets processed per second, and memory footprint and computational resource utilization.
- **Analyse the trade-offs** between model size, embedding dimensions, and real-world deployment feasibility across different use case scenarios, providing evidence-based recommendations for organizations making architectural decisions about their ITSM AI infrastructure.
- **Develop a containerized deployment architecture** that delivers a vendor-independent alternative to ServiceNow's proprietary AI solutions, enabling organizations to deploy advanced ticket classification capabilities on local servers or cloud-hosted container platforms (AWS, Azure, GCP, Kubernetes) without vendor lock-in or recurring AI licensing costs.
- **Deliver a working prototype with full ServiceNow integration**, comprehensive documentation, and an interactive dashboard that visualizes comparative results and provides actionable insights to consultants.
- **Explore the feasibility of detecting causal or dependent relationships** between incident tickets beyond simple similarity matching, investigating whether patterns in ticket descriptions and metadata can reveal deeper connections that point toward shared root causes or cascading failures.

Industry Relevancy:
(Please use a separate sheet if necessary)

Please provide information regarding the relevance of the project to the hosting organization, and / or to the industry at large

Vendor-Independent AI for ITSM
Organizations using ServiceNow for IT Service Management often face a critical decision: adopt ServiceNow's proprietary Predictive Intelligence (with associated licensing costs and vendor lock-in) or forgo AI-powered ticket classification entirely. This project delivers a third option: a containerized, standalone AI classification service that integrates seamlessly with ServiceNow via API while remaining completely independent of ServiceNow's AI ecosystem.

By packaging the solution as deployable Docker containers, organizations gain unprecedented flexibility:

- **Deploy anywhere**: Local servers, AWS, Azure, Google Cloud, on-premises Kubernetes clusters, or hybrid environments based on security, compliance, and cost

considerations
- **Avoid vendor lock-in**: Maintain control over AI infrastructure and model selection without dependency on proprietary systems
- **Cost control**: Eliminate recurring AI licensing fees while achieving comparable or superior classification accuracy
- **Customization freedom**: Fine-tune open-source embedding models for organization-specific terminology and ticket patterns
- **Transparency**: Full visibility into model behaviour and decision-making processes, unlike black-box proprietary solutions
- **Portability**: Architecture can potentially adapt to other ITSM platforms (Jira Service Management, Zendesk, Freshservice) beyond ServiceNow

Operational Efficiency and Business Impact

In large enterprises, IT Service Management (ITSM) systems handle thousands of incident tickets daily across multiple applications, modules, and business units. These incidents are often repetitive or related to the same root cause but are logged as new tickets, leading to duplicate analysis, inconsistent resolutions, and extended downtime. Functional consultants spend significant time triaging and categorising similar issues, which diverts attention from higher-value work.

Implementing an automated categorisation and ticket-linking system addresses these inefficiencies directly. By enabling the system to recognise patterns in incident descriptions and automatically associate them with known cases, organisations can:

- Reduce manual triage and reassignment cycles
- Improve consistency in problem resolution by referencing historical knowledge
- Lower Mean Time to Resolution (MTTR) through faster identification of root causes
- Strengthen knowledge management by building a searchable, context-rich database of incident clusters

For service providers or multinational organisations, this translates to tangible benefits: operational efficiency, reduced support costs, and improved service-level compliance.

Strategic Model Selection and Deployment Architecture
The choice of embedding model has significant practical implications for deployment architecture and operational costs. Larger models like Qwen3-embedding-8B may offer superior accuracy but require substantial computational resources and longer inference times, while smaller models like all-MiniLM-L6-v2 provide faster responses suitable for high-volume production environments with thousands of daily tickets. Mid-range options like Gamma-embedding-300M and all-mpnet-base-v2 offer different balances between these extremes.

By systematically comparing models across different dimensions and parameter counts, this project provides evidence-based guidance for organizations making architectural decisions about their ITSM AI infrastructure. The containerized deployment package enables organizations to easily switch between models based on their specific needs—deploying lightweight models for real-time classification or larger models for batch processing scenarios where accuracy is paramount.

Understanding these trade-offs enables better cost-benefit analysis when choosing between on-premises deployment, cloud-based inference, or hybrid approaches, directly impacting both service quality and operational expenses. Organizations can start with one model configuration and seamlessly switch to another as requirements evolve, without vendor negotiations or platform migrations.

Project Scope:
(Please use a separate sheet if necessary)

Please provide a detailed description of the project scope, including any limitations of the project

The project focuses on developing and evaluating an AI-driven approach for incident ticket categorization and linking within IT Service Management (ITSM) environments. Using either publicly available ServiceNow-format sample data or desensitized internal datasets, the system will apply Natural Language Processing (NLP) techniques to classify incoming tickets and identify relationships with past incidents.

Core Components:

1. **ServiceNow API Integration Pipeline:**
    a. Automated extraction of incident ticket data including descriptions, metadata, and historical categories
    b. Processing through custom embedding models with real-time or batch capabilities
    c. Automated pushing of classification results, confidence scores, and linkage suggestions back to ServiceNow
2. **Multi-Model Comparison Framework:**
    a. **all-MiniLM-L6-v2**: Lightweight model (384 dimensions, ~80MB) optimized for speed, suitable for high-throughput real-time classification scenarios
    b. **Gamma-embedding-300M**: Mid-size model (300M parameters, ~229MB) offering balanced performance for general production use
    c. **all-mpnet-base-v2**: Standard baseline (768 dimensions, ~420MB) providing quality-focused embeddings with proven reliability
    d. **Qwen3-embedding-8B**: Large-scale model (8B parameters, ~4.68GB) for maximum accuracy in scenarios where classification quality is critical
3. **Performance Metrics and Evaluation:**
    a. Classification accuracy measured across different ticket categories and complexity levels
    b. Linking precision and recall for identifying related, duplicate, or dependent tickets
    c. Inference speed measured in tickets processed per second under realistic load conditions
    d. Memory footprint and computational resource requirements for each model
    e. Analysis of accuracy-speed tradeoffs to inform deployment decisions
4. **Relationship Detection Capabilities:**
    a. The system will recommend ticket groupings or parent-child linkages based on textual similarity and contextual features extracted from ticket descriptions.
    b. Additionally, the project will **explore the feasibility of detecting deeper or causal relationships** between tickets---such as when one incident may contribute to or result from another. This exploratory component aims to identify patterns suggesting root-cause relationships that could guide more strategic incident resolution.
5. **Interactive Visualization Dashboard:**
    a. A user-friendly interface to visualize classification results, linked or related

tickets, identified dependencies, and comparative model performance metrics. This dashboard provides consultants with a clearer view of issue patterns, potential root causes, and helps them understand which model offers the best balance for their specific operational requirements.

6. **Containerized Deployment System (Vendor-Independent Architecture):**
    a. Docker containers packaged as standalone microservices independent of ServiceNow's AI ecosystem
    b. Docker Compose orchestration for one-command deployment on local servers or cloud platforms
    c. Cloud-agnostic architecture supporting AWS ECS, Azure Container Instances, Google Cloud Run, and Kubernetes
    d. Configuration management enabling seamless switching between embedding models based on operational requirements
    e. Volume management for model persistence, ticket data caching, and result storage
    f. Environment-based configuration for ServiceNow API integration without platform dependencies
    g. Resource allocation profiles optimized for different deployment scenarios (on-premises, cloud, hybrid)
    h. Pre-built container images enabling rapid deployment without vendor-specific tooling or proprietary platforms
7. **Limitations:**
    a. Causal-link detection will remain exploratory and dependent on dataset quality and available metadata
    b. All analysis will be conducted using anonymized or sample data to ensure privacy compliance
    c. Model selection and deployment recommendations will balance accuracy requirements with practical deployment constraints (memory, inference speed, cost)
    d. API integration assumes standard ServiceNow REST API access and appropriate permissions
    e. Performance benchmarks are specific to the hardware and dataset used; results may vary in different production environments

Project Outputs and Deliverables
(Please use a separate sheet if necessary)

Please provide a detailed description of the expected outputs and deliverables by the candidate

1. **ServiceNow Integration Pipeline:**
    a. Functional API-based data extraction module with error handling and retry logic
    b. Automated result update system for pushing classifications, confidence scores, and linkage suggestions back to ServiceNow
    c. Integration documentation and deployment guide
2. **Four Fine-Tuned Embedding Models:**
    a. all-MiniLM-L6-v2 implementation with comprehensive performance benchmarks
    b. Gamma-embedding-300M implementation with comprehensive performance benchmarks
    c. all-mpnet-base-v2 implementation with comprehensive performance benchmarks

d. Qwen3-embedding-8B implementation with comprehensive performance benchmarks
e. Model training procedures, hyperparameters, and fine-tuning methodology documentation

3. **Comparative Evaluation Report:**
    a. Detailed analysis of accuracy, precision, recall, inference speed, and memory trade-offs across all four models
    b. Statistical significance testing of performance differences between models
    c. Deployment recommendations based on different use-case requirements (high-volume vs. high-accuracy scenarios)
    d. Cost-benefit analysis framework for model selection decisions
    e. Visualization of performance trade-offs across multiple dimensions
4. **Interactive Dashboard:**
    a. Real-time visualization of ticket classifications with confidence scores
    b. Display of linked and related ticket clusters with similarity metrics
    c. Model performance comparison interface showing accuracy-speed tradeoffs
    d. Exploratory visualization of potential causal relationships between tickets
    e. User-friendly interface for consultants to explore ticket patterns and relationships
5. **Exploratory Analysis on Causal Relationships:**
    a. Findings on potential causal or dependent ticket relationships detected by the models
    b. Analysis of patterns suggesting root-cause connections between incidents
    c. Discussion of limitations and confidence levels in causal relationship detection
    d. Future research directions for improving causal inference in ITSM contexts
6. **Technical Documentation:**
    a. System architecture diagrams and pipeline design specifications
    b. Model training procedures, data preprocessing steps, and evaluation methodologies
    c. ServiceNow API integration guide with code examples
    d. Deployment recommendations for different organizational scales and requirements
    e. Troubleshooting guide and maintenance procedures
7. **Final Presentation:**
    a. Executive summary of results, key findings, and practical implications
    b. Detailed comparison of model performance with visual aids
    c. Live demonstration of the complete end-to-end system
    d. Discussion of limitations, lessons learned, and future work opportunities
8. **Containerized Deployment Package:**
    a. Docker files for all system components (API integration service, model inference engine, interactive dashboard)
    b. Docker Compose configuration files for one-command deployment and orchestration
    c. Pre-built container images for all four embedding models with pull-and-run capability d. Configuration templates for ServiceNow integration with secure credential management

d. Resource requirement documentation specifying CPU, memory, and storage needs for each model configuration
e. Quick-start deployment guide with prerequisites, installation steps, and troubleshooting procedures
f. Model switching scripts and system health monitoring utilities
g. Container optimization documentation for production deployment scenarios on local and cloud infrastructure

C. Knowledge and Training Requirements

Applicable Knowledge from the Degree Programme

(Please use a separate sheet if necessary)

Please provide details of any pre-requisite knowledge and / or skillsets from the degree programme that is required by the candidate for the project

- Data preprocessing and database management.
- Machine Learning and Deep Learning fundamentals.
- Natural Language Processing concepts.
- Software development and systems integration.

Additional Knowledge, Skillsets, or Certifications Required

(Please use a separate sheet if necessary)

Please provide details of any other required knowledge, skillsets, or certifications beyond the degree programme that the candidate will require to complete the project

- Experience with ServiceNow platform and Predictive Intelligence configuration.
- Knowledge of NLP frameworks (e.g., Hugging Face Transformers, Sentence-Transformers).
- Familiarity with model evaluation techniques and experiment tracking.
- Docker and containerization fundamentals for application packaging and deployment.
- Docker Compose for multi-container orchestration and service management.
- Resource management and optimization strategies for ML models in containerized environments.

Training Required and Provided

(Please use a separate sheet if necessary)

Please provide details of any additional relevant training that is required by, and that will be provided to the candidate for the purpose of the project

- Access and export procedures for ServiceNow data.
- Hands-on learning in text similarity and classification modelling.
- Optional training in cloud-based model execution and cost management.

D. Project Timeline

(Please use a separate sheet if necessary; a Gantt chart may be included as annex)

Please provide a detailed timeline of the project, including expected output and deliverables

Phase 1 – Data Preparation and Analysis
Obtain ServiceNow-format dataset (sample or anonymized). Ensure data quality and perform comprehensive preprocessing, cleaning, and exploratory analysis to understand text structures, category distributions, ticket volume patterns, and potential relationships between incidents. Establish baseline statistics and identify data quality issues that may affect model training.

Phase 2 – Embedding Model Comparison Framework Setup
Establish a standardized evaluation framework for fair comparison across all four embedding models. Set up infrastructure for training, testing, and benchmarking with consistent hardware and environment configurations. Prepare identical train/validation/test splits ensuring no data leakage. Define and implement evaluation metrics (accuracy, precision, recall, speed, memory) with automated measurement scripts.

Phase 3 – Multi-Model Training and Evaluation
Train and fine-tune all four embedding models (all-MiniLM-L6-v2, Gamma-embedding-300M, all-mpnet-base-v2, Qwen3-embedding-8B) using the prepared dataset. Optimize each model for ticket classification and similarity matching while carefully documenting training procedures, hyperparameters, convergence behaviour, and any challenges encountered. Conduct initial performance assessments to validate training effectiveness.

Phase 4 – Performance Analysis and Model Selection
Evaluate all models on identical test data using the standardized metrics framework. Conduct comprehensive comparison of classification accuracy, linking precision/recall, inference speed, and memory usage. Perform statistical significance testing to validate performance differences. Analyse trade-offs and determine optimal model recommendations for different deployment scenarios (high-volume vs. high-accuracy, real-time vs. batch processing).

Phase 5 -- Pipeline Integration, Containerization, and Dashboard Development
Implement full ServiceNow API integration including data extraction and automated result updates. Develop containerized deployment architecture using Docker and Docker Compose, packaging all four embedding models as swappable container configurations for rapid switching between performance profiles. Create cloud-agnostic deployment templates supporting local servers, AWS, Azure, GCP, and Kubernetes environments. Build the interactive dashboard to visualize classifications, linked tickets, model performance comparisons, and exploratory causal relationships. Test the end-to-end pipeline functionality with realistic workloads and validate containerized deployment across different environments. Refine the user interface based on usability considerations and consultant feedback.

Phase 6 – Documentation and Final Presentation
Compile comprehensive technical documentation covering system architecture, methodology, evaluation results, and deployment guidance. Prepare the final report synthesizing evaluation findings, trade-off analysis, deployment recommendations, and

lessons learned. Develop the final presentation with clear visualizations and prepare for live system demonstration.

E. Supervisor Remarks

(Additional details relevant to the project; please use a separate sheet if necessary)

E.1 Industry Supervisor Remarks

Project/ Assignment aim to improve Service Now usage with AI as outlined in project objective and achieve some of below improvements

Incident Management

- **Automated Ticket Categorization**: AI can classify incidents based on historical data and context, reducing manual effort.
- **Predictive Assignment**: Suggests the best resolver group or agent based on past resolution patterns.
- **Anomaly Detection**: Identifies unusual spikes or patterns in incident volumes that may indicate systemic issues.

. Performance Analytics & Reporting

- **Predictive Analytics**: Forecasts future trends in service demand or incident volumes.
- **Root Cause Analysis**: Uses machine learning to identify underlying causes of recurring issues.
- **Sentiment Analysis**: Evaluates user feedback to improve service quality.

Knowledge Management

- **Intelligent Search**: AI improves search relevance by understanding context and user behaviour.
- **Auto-Knowledge Article Creation**: Suggests or drafts articles based on resolved incidents.
- **Content Gap Identification**: Detects missing or outdated knowledge content.

Customer Service Management

- **Case Deflection**: AI suggests relevant knowledge articles to customers before they submit a case.
- **Customer Sentiment Analysis**: Helps prioritize cases based on emotional tone and urgency.

E.2 Academic Supervisor Remarks

I suggest putting also put emphasis on the visualisation of the ticket-linking / relationship aspects, going beyond just related / duplicate tickets to also causal relationships (one issue being the cause of another one).

F. Declaration of Conformity

F.1 Declaration by Industry Supervisor

I hereby acknowledge that I have engaged and discussed with the Candidate on the Capstone Project details above, and we are in agreement on the contents provided in the sections above.

Signature Spreddy

Name of Industry Supervisor: Praveen
Designation: SAP Application Delivery Management Lead, Application Managed Service
Department: Managed Services ADM/AMS, PISCAP
Contact No: 96501275
Email: praveen.reddysatti@sg.panasonic.com
Date: 13/10/2025

F.2 Declaration by Candidate

I hereby acknowledge that I have engaged and discussed with my Industry Supervisor on the Capstone Project details above, and we are in agreement on the contents provided in the sections above.

Signature

Name of Candidate: Donovan Leong Jia Le
Contact No: 91913435
Email: 2302674@sit.singaporetech.edu.sg
Date: 12/10/2025

F.3 Declaration by Academic Supervisor

I hereby acknowledge that I have read and understood the Capstone Project details above. I believe that the proposed project is appropriate to fulfil the requirements for the **AAI4001 Capstone Project** module.

Signature

Name of Academic Supervisor: Bogdan Cautis
Contact No: 8229 0321
Email: bogdan.cautis@singaporetech.edu.sg
Date: 12/10/2025

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